

MATLAB: Session 9

Vincenzo Antonio Isoldi
Francesco Giuseppe De Vitis

27 November 2025

Exercises

Part A – Eigenvalues, Eigenvectors, Diagonalizability

A.1 Consider the endomorphisms $g_k : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ defined as

$$g_k(x, y, z, w) = (ky + w, x - w, x - ky + z - w, w),$$

where k is a real parameter.

Assign to the variable **Bk** the matrix representing g_k with respect to the canonical basis of $\mathbb{R}^4$.

A.2 Compute the characteristic polynomial of g_k and assign it to **chargk**.

A.3 Compute the eigenvalues of g_k and assign them to the column vector **eiggk** (eigenvalues with algebraic multiplicity greater than one, must be repeated).

A.4 Compute the eigenvectors of g_k . Save them as follows: if $v_1, \dots, v_j$ form a basis for the eigenspace relative to the eigenvalue $\lambda = a$, then assign them as columns of the matrix **Vgk_a**.

A.5 For which k 's is g diagonalizable?

A.6 For $k = 1$, assign the diagonal form of **B1** to the variable **Diagg1**.

A.7 In the latter case, compute the diagonalizing matrix and assign it to **Qg1**.

Part B – Spectral Theorem

B.1 Consider the endomorphism $s : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, defined as

$$s(x, y, z) = (x + 2z, -y, 2x + z).$$

Assign to the variable **E** the matrix representing s with respect to the canonical basis of $\mathbb{R}^3$.

B.2 Without computing anything: is s diagonalizable?

B.3 Compute the characteristic polynomial of s and assign it to **chars**.

B.4 Compute the eigenvalues of s and assign them to the column vector **eigs** (eigenvalues with algebraic multiplicity greater than one, must be repeated).

B.5 Compute the eigenvectors of s . Save them as follows: if $v_1, \dots, v_j$ form a basis for the eigenspace relative to the eigenvalue $\lambda = a$, then assign them as columns of the matrix **Vs_a**.

B.6 Consider the eigenspaces computed in the previous point. Do they have any specific properties?